

HAT-P-19b

R-1218

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-19_231021 · created 2026-07-31T08:33:09+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460238.8201 +/- 0.005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.17 +/- 0.053
Transit depth (Rp/Rs)^2	2.88 +/- 1.8 [%]
Orbital Inclination (inc)	86.01 +/- 1.6
Airmass coefficient 1 (a1)	1.27 +/- 0.62
Airmass coefficient 2 (a2)	-0.2 +/- 0.32
Scatter in the residuals of the lightcurve fit is	50.14 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	7.5
Transit Duration (day)	0.06 +/- 0.046

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.17 ± 0.053 vs 0.1418 (deviation 0.37σ)
Duration fit vs published	0.06 d vs 0.1196 d (deviation 0.9σ)
Mid-time O–C	10.5 min
Depth SNR	2.2
Residual scatter	50.14 %

Light curve

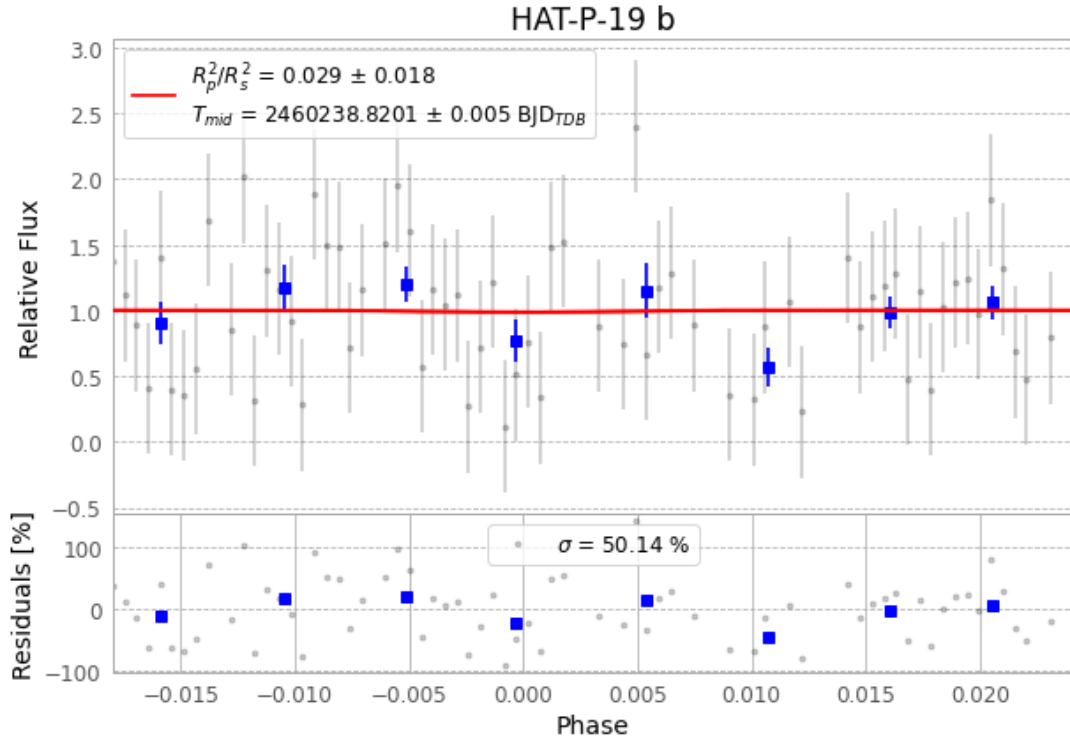


Figure 1 — FinalLightCurve_HAT-P-19 b_2023-10-20.png (SHA-256 0461e79bc76fc68f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [191, 439], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 d7ad9ce755b669b2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

82 FITS frames (54 MB), HATP-19231021054815.FITS → HATP-19231021095116.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-231021.log (c3b3b00ecc41...), FinalLightCurve_HAT-P-19 b_2023-10-20.png (0461e79bc76f...), FinalLightCurve_HAT-P-19 b_2023-10-20.csv (4b44e784260f...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 08:33Z.